

Universal Pre-training for Images

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Abstract

This thesis investigates whether Vision Transformers can be pre-trained from image-like data generated by iterated random computation, without using real images during pre-training. The motivation comes from universal pre-training: pure randomness is not useful for learning, but randomness transformed by computation can contain structure. We implement an IRC image generator based on a persistent buffer and randomly sampled convolutional updates, and train a ViT-Tiny encoder with a masked-autoencoder objective on the generated stream. The frozen encoder is then evaluated on real and synthetic downstream datasets using linear probing, together with zero-adaptation reconstruction metrics. We compare IRC pre-training against random initialization, Gaussian blur noise, Fourier textures, and a compact real-video baseline based on a single walking video. The results show that IRC pre-training consistently improves over random initialization and over simpler synthetic baselines. More importantly, on the classification tasks, IRC also outperforms the real-video baseline, even though the video baseline uses real visual data. The method does not replace large-scale real-image pre-training, and it does not yet learn strong object semantics. However, the results provide evidence that useful visual structure can be learned before the model sees real images, supporting the broader idea that part of visual pre-training may come from general computational structure rather than from real-image data alone.

1 Introduction

Modern deep learning has benefited enormously from scale. Larger models, trained on larger datasets, often learn representations that transfer better, converge faster, and generalize more robustly to downstream tasks [1, 2, 3]. In computer vision, this has led to a standard recipe: first pre-train an image encoder on a large generic dataset, and then fine-tune it on a smaller target dataset [4, 5, 2]. This pipeline has been central to the success of modern visual representation learning, especially for Vision Transformers (ViTs), which process images as sequences of patch tokens and inherit much of their design from the Transformer architecture used in sequence modeling [6, 3]. However, this recipe also hides a strong assumption: we must have access to a large collection of real images.

This assumption is not always realistic. In many domains, data is expensive, restricted, private, or simply unavailable at the required scale. One domain in which this difficulty is particularly acute is medical imaging. Medical images are often protected by privacy regulations, collected under heterogeneous acquisition protocols, and tied to sensitive patient information [7]. Even when images can be accessed, labels may require expert annotation, and for rare diseases the number of positive examples may be intrinsically small [7]. Moreover, medical images can vary across hospitals, scanners, acquisition settings, and patient populations, making small datasets even harder to use

reliably [8]. These constraints motivate a central question for data-efficient vision: can we reduce the dependence of visual pre-training on large real-image datasets?

Self-supervised learning removes one important bottleneck, namely the need for manual labels, but it does not remove the need for images [9, 10, 11]. A self-supervised visual encoder still requires a large pool of real visual observations from which to extract structure [12, 13]. Thus, the scarcity problem is not only a labeling problem. It is also a data-availability problem. If the target domain is private, specialized, or rare, then even unlabeled data may be limited. In such settings, a useful initialization can be as valuable as an additional dataset: it may reduce the amount of real domain data required to reach a given level of performance [14, 8].

The difficulty becomes clearer when considering the size of image space. An RGB image of height H and width W , with 8-bit color channels, belongs to a space of

$$256^{H \cdot W \cdot 3}$$

possible arrays. Even for a relatively standard ViT input resolution of 224×224 , this is approximately $10^{362,500}$ possible images. For a 1024×1024 image, which is still modest compared with some high-resolution medical and scientific images, the number is approximately $10^{7,575,700}$. By comparison, the number of atoms in the observable universe is often estimated to be on the order of 10^{80} [15]. The raw image space is therefore not merely large; it is overwhelmingly larger than any dataset we can hope to collect.

Yet natural images occupy only a tiny and highly structured subset of this space [16]. Almost every possible RGB array is meaningless noise. Useful images, in contrast, contain regularities: neighboring pixels are often correlated, edges form contours, contours enclose regions, textures repeat, objects have boundaries, shapes compose into scenes, and distant patches can depend on one another [17, 18, 19]. Vision is therefore not only about observing pixels, but about learning the structure that makes some pixel arrays meaningful and most others useless. This leads to the question at the core of this work: can a model learn useful visual structure without seeing any real images?

At first glance, this may seem impossible. If synthetic data is generated by a process we design ourselves, it is natural to ask where the useful information could come from. In information-theoretic terms, computation cannot create new information from nothing [20, 21]. However, useful training data is not valuable merely because it contains many bits. Pure random noise has maximal unpredictability, but almost no learnable structure. What matters for pre-training is the combination of variation and regularity: randomness provides diversity, while computation can turn that randomness into dependencies, repetitions, hierarchy, and constraints [21].

A useful intuition is to compare two kinds of randomness. In the first case, we sample every pixel of an image independently. The result is similar to television static: it contains many bits, but almost no exploitable structure. In the second case, we first sample random pixels and then pass them through a randomly chosen computational process. This process might blur local neighborhoods, copy patches, impose symmetries, threshold intensities, warp coordinates, apply filters, or repeatedly feed intermediate outputs back into itself. Such computation does not add information in the Shannon sense, and it does not inject real-world semantics [20, 21]. However, it can transform independent randomness into structured randomness. The resulting images may not depict real objects, but they can contain edges, regions, textures, repeated patterns, spatial continuity, and dependencies between distant patches. These are precisely the kinds of regularities from which a vision model can begin to learn. In this sense, random computation can act as a bridge between meaningless noise and useful visual structure.

This perspective is closely related to *universal pre-training*. Instead of first collecting data from the target domain, universal pre-training asks whether a model can be pre-trained on data

generated before the downstream task is known. In the sequence setting, *Universal pre-training by iterated random computation* proposes to generate structured random data by passing randomness through random computational processes, and to use this data to pre-train a model that can later adapt to many different tasks [21]. The most important idea is not that the generated data contains the semantics of future tasks. Rather, the hope is that many real-world data sources are themselves produced by computational or physical processes, and therefore share broad structural properties. Generator-specific patterns should average out across many random processes, while more general patterns may remain useful [21].

The same view naturally suggests a data–computation tradeoff. When real data is abundant, a model can learn structure directly from examples. When real data is scarce, private, or expensive, we may instead spend computation to generate structured pre-training signals [21]. Such pre-training can then be amortized over many downstream tasks: the expensive synthetic pre-training stage is performed once, while the resulting initialization can be reused whenever real target data is limited. This is especially attractive for domains such as medical imaging, where obtaining an additional large dataset may be harder than spending additional computation.

Vision Transformers provide a particularly natural testbed for this idea. A ViT converts an image into a sequence of patches and processes these patches with a Transformer encoder [6, 3]. This patch-sequence formulation creates a bridge between sequence learning and image learning: the input is visual, but the architecture reasons over tokens and their relations. At the same time, ViTs contain fewer image-specific inductive biases than convolutional networks. Locality, translation equivariance, and two-dimensional neighborhood structure are not built into every layer in the same way as in CNNs; much of the spatial structure must be learned from data [3]. This makes ViTs powerful, but also data-hungry: large ViTs tend to benefit most from large-scale pre-training, while smaller or synthetic pre-training regimes require careful design to compensate for the weaker built-in visual priors [3, 22, 23].

There is already evidence that useful visual representations need not come exclusively from natural photographs. Prior work has shown that models can learn from procedural noise processes, provided that the generated images contain suitable structural properties such as naturalistic statistics and sufficient diversity [16]. Formula-driven supervised learning and fractal-based datasets have also shown that visual encoders can be pre-trained on synthetic images generated from mathematical rules rather than real image collections [24, 25]. In particular, one-instance fractal datasets suggest that even a very small number of synthesized images can provide useful ViT pre-training when combined with appropriate augmentation [23]. More broadly, generated image–caption pairs have also been shown to support strong visual representation learning without directly training on real images [26]. These results indicate that realism itself may not be the essential ingredient. A synthetic image does not necessarily need to look like a photograph in order to teach a model something useful about visual organization.

Recent work pushes this idea even further. Procedural warm-up for Vision Transformers shows that non-visual symbolic data, generated from formal grammars and used before image-based training, can improve convergence and downstream performance [27]. This suggests that some helpful priors for vision may be computational rather than purely visual. A model may benefit from learning generic mechanisms such as dependency tracking, hierarchy, composition, and long-range interaction, even before it is exposed to pixels.

Together, these observations point to a gap. Synthetic images, fractals, structured noise, generated image–caption pairs, and symbolic procedural data have all been explored as alternatives or complements to real visual data [16, 24, 23, 26, 27]. Separately, universal pre-training has been studied for token sequences and explicitly proposed as a direction that could, in principle, extend to computer vision [21]. However, it remains unclear how universal random computation should

be adapted directly to image-based ViT pre-training. In particular, the vision setting raises new questions: what kind of generated image-like structure is useful, how should randomness and spatial regularity be balanced, and which visual priors can be learned before the model sees real images?

This work is motivated by the following question: can a Vision Transformer learn useful visual representations from universally generated synthetic image-like data before seeing any real images? The goal is not to replace domain-specific data entirely. Real data remains necessary for evaluation, adaptation, and learning task-specific semantics. Instead, the goal is to reduce the amount of real data required by starting from a more useful initialization. More broadly, this paper studies how much of “learning to see” depends on real images, and how much can be acquired from structure alone.

1.1 Research Question

The main research question of this thesis is:

Can Vision Transformers acquire transferable visual representations from image-like data generated by iterated random computation, without using real images during pre-training?

2 Related Work

A useful way to view the literature is as a gradual movement away from the classical assumption that visual pre-training must start from a large, labeled, real-image dataset. The earliest successful recipe was simple and powerful: collect a large image dataset, train a model on it, and transfer the learned representation to smaller tasks. ImageNet played a central role in this story, not only as a benchmark, but also as a reusable source of visual features [4, 5]. Later, larger datasets such as JFT and large-scale transfer methods showed that increasing the amount of pre-training data could further improve downstream performance [1, 2]. In this view, the model learns from the world because the dataset already contains a large and diverse sample of the world.

Vision Transformers made this dependence on data especially visible. By representing images as sequences of patches, ViTs brought the Transformer architecture from language into vision [6, 3]. This made the architecture simple and scalable, but also reduced some of the image-specific inductive biases that convolutional networks have by design. Locality, translation equivariance, and two-dimensional neighborhood structure are not built into ViTs in the same strong way as in CNNs; instead, many spatial relations must be learned from data [3]. This helps explain why large-scale pre-training is so important for ViTs, and why data-efficient ViT training has become an important research direction [22, 23].

A first step away from labeled data is self-supervised learning. Methods such as SimCLR, MoCo, DINO, MAE, and iBOT show that strong visual representations can be learned without manual labels, by designing learning objectives from the images themselves [28, 29, 9, 10, 11]. More recent foundation models such as DINOv2 and DINOv3 scale this idea further and demonstrate that self-supervised image encoders can become broadly useful across classification, dense prediction, and other vision tasks [12, 13]. However, these methods still depend on large collections of real images. They remove the label bottleneck, but not the image bottleneck. For domains where even unlabeled data is scarce or private, this distinction matters.

Another line of work asks whether we can use real data more efficiently. DeiT showed that training choices such as distillation and augmentation can make ViTs more data-efficient [22]. More radically, DORA investigates whether a strong image encoder can be learned from a single

long unlabeled video, rather than from a large collection of independent images [30]. This result is important for the present thesis because it suggests that the organization of data can matter as much as its raw size. A continuous video contains temporal coherence, object persistence, viewpoint changes, and natural transformations. In other words, one stream can contain many reusable regularities. This supports the broader idea that useful pre-training may come not only from having many examples, but from exposing the model to the right kind of structure.

A more direct challenge to real-image pre-training comes from synthetic and procedural visual data. FractalDB and formula-driven supervised learning show that models can be pre-trained on images generated from mathematical rules rather than collected from cameras [24]. Follow-up work on one-instance fractal databases shows that even a very small number of synthesized images can be useful for ViT pre-training when combined with suitable augmentation [23]. These results weaken the assumption that visual pre-training requires natural images. They suggest that synthetic images may provide enough geometric and spatial variation for a model to learn transferable features.

This idea is also developed in work on learning from noise. Baradad et al. study several procedural noise processes and show that visual representations can be learned from generated images that are not natural photographs [16]. Their main message is especially relevant here: the best synthetic data is not necessarily the most realistic data. Instead, useful generated data should balance diversity with structural properties similar to those found in natural images. Procedural Image Programs extend this direction by using richer procedural generators as a source of visual pre-training data [31]. Together, these works suggest that realism is not the only path to useful representation learning. What matters is that the generated images contain learnable visual regularities.

A different approach is to use generative models as data sources. SynCLR, for example, uses language models to synthesize captions and text-to-image models to synthesize images, then trains visual representations on the resulting generated image-caption pairs [26]. This shows that a visual encoder can be trained without directly using a real-image dataset as its pre-training set. However, the source of structure in this case still comes from powerful generative models that were themselves trained on large real-world corpora. This makes SynCLR very effective, but conceptually different from the setting considered in this thesis. Here, the goal is not to distill visual knowledge from already-trained generative models, but to ask whether useful structure can arise from random computation itself.

The most closely related conceptual starting point is universal pre-training. Universal pre-training studies whether a model can be trained before the downstream data is known, using synthetic data generated from random computational processes [21]. The key idea is that pure randomness is not useful, but randomness passed through computation can become structured. Such data may contain dependencies, repetition, hierarchy, and regularities that are not specific to one dataset. If many real-world data sources are produced by computational or physical processes, then a model trained on many random computational sources may learn patterns that are broadly reusable. Universal pre-training therefore frames pre-training as a data-computation tradeoff: when real data is limited, one may spend computation to generate useful structure [21].

Finally, Procedural Warm-Up for Vision Transformers connects this idea directly to vision models. Instead of using images, it pre-trains ViTs on symbolic sequences generated by formal grammars before standard image training [27]. The surprising result is that non-visual procedural data can improve convergence and downstream performance. This suggests that some useful priors for vision may be computational rather than purely visual: hierarchy, dependency tracking, composition, and long-range interaction may help a Transformer before it ever sees pixels.

Taken together, prior work has moved through several stages: from labeled real images, to unlabeled real images, to smaller or more structured real data, to synthetic images, to non-visual

procedural data. This thesis continues that trajectory. Unlike synthetic-image methods that hand-design visual generators, and unlike procedural warm-up methods that bypass the image domain, this work asks whether iterated random computation can be adapted directly to image-like pre-training data for Vision Transformers. The central gap is therefore not whether synthetic data can help vision models in general, but whether universal random computation can generate visual structure that is useful before any real images are observed.

3 Approximating Universal Pre-training for Images

The previous sections motivate the main idea of this thesis: useful pre-training data does not necessarily have to be real, but it should contain learnable structure. Universal pre-training gives a general version of this idea. In principle, one would like to sample data from a very broad distribution over computable sources, so that a model can learn patterns that are not tied to one specific dataset or task [21]. In practice, however, the true universal distribution cannot be sampled directly. We cannot enumerate all programs, we cannot run them for unbounded time, and we cannot store an infinite stream of generated data.

The method used in this thesis is therefore a practical approximation. It does not claim to sample from the true universal distribution. Instead, it tries to capture one useful principle behind it: start from randomness, repeatedly apply randomly sampled computations, and train a model on the structure that appears. In the image setting, this means maintaining a buffer of random image-like tensors and repeatedly transforming them with randomly initialized convolutional networks. The generated images are not designed to look like natural photographs. Their role is more basic: they should be structured enough for a Vision Transformer to learn spatial and computational regularities.

3.1 From Random Computation to Image-like Data

Universal pre-training was originally studied in a sequence setting, where random bits are passed through random computations to produce structured token sequences [21]. Images are different from text, but Vision Transformers make the connection natural. A ViT converts an image into a sequence of patch tokens and processes those tokens with a Transformer encoder [6, 3]. Therefore, the vision version of the problem can be phrased as follows: can we generate image-like tensors whose patch sequences contain useful structure?

Let an image be represented as

$$x \in [0, 1]^{3 \times R \times R},$$

where R is the image resolution. In the final experiments, $R = 224$. The generator starts from random images in this space and repeatedly applies random image computations. The family of computations used here is intentionally simple: small randomly initialized convolutional networks. This is not as general as all possible programs, but it is a reasonable first step for images. Convolutions preserve the spatial layout of the image, create local dependencies, and can be applied repeatedly to increase the effective computational depth.

This choice also keeps the method conceptually close to the motivation from the introduction. Pure pixel noise contains many bits, but almost no predictable organization. After random convolutional processing, the image may contain smooth regions, color correlations, edges, textures, repeated motifs, and larger spatial dependencies. These are not semantic objects, but they are still forms of structure. Prior work on procedural noise also suggests that realism is not always the crucial property of synthetic data; diversity and useful image-like regularities can matter more [16].

3.2 The IRC-Convolutional Generator

The main generator used in this thesis is called `irc_conv`. The name stands for iterated random computation with convolutional image updates. The generator keeps a persistent buffer

$$\mathcal{B}^{(t)} = \{x_1^{(t)}, x_2^{(t)}, \dots, x_B^{(t)}\},$$

where each $x_i^{(t)}$ is a generated image at training step t . In the final run, the buffer size is $B = 20,000$.

The buffer is initialized from independent RGB noise. In addition, 10% of the initial buffer is replaced by random solid-color images. These solid-color images are simple low-frequency seeds: they do not contain object information, but they make it easier for later random computations to create smooth regions and color structure.

At every training step, only part of the buffer is refreshed. In the final configuration, the refresh fraction is 2.5%, so 500 out of 20,000 buffer images are selected for update. A fresh random convolutional network is then sampled for the batch. In the final run the random network is not fixed for the whole experiment. Instead, each generator call uses a newly sampled random computation. This is important because the aim is not to learn from the artifacts of one generator, but from many different computational sources.

The random convolutional network has width 16, depth 3, kernel size 3, dilation 1, and GELU hidden activations. Its weights are sampled from a zero-mean Gaussian distribution with standard deviation 0.08, and the weights are never trained. Given a selected buffer image x , the network produces an update $g_\phi(x)$, where ϕ denotes the random weights. The output is passed through a sigmoid and blended with the current image:

$$x \leftarrow Q((1 - \alpha)x + \alpha \sigma(g_\phi(x))).$$

Here, σ is the sigmoid function, α is the update strength, and Q denotes clipping to $[0, 1]$ followed by optional quantization to the 8-bit image grid. In the final configuration, quantization is enabled, so the generated images lie on the same 256-level-per-channel grid as standard RGB images.

Each refreshed image receives $k = 2$ random convolutional updates. The update strength is scheduled during training: it starts at $\alpha = 0.30$ and decreases linearly to $\alpha = 0.15$ over the first 10 million generated images. The intuition is that stronger updates early in training move the buffer away from independent noise, while weaker updates later in training preserve more of the structure that has accumulated.

The generator also injects fresh sources of variation. A small fraction of the buffer is reset to independent noise. This reset fraction decreases linearly from 0.1% to 0.01% over the first 10 million generated images. With a buffer of 20,000, this corresponds to approximately 20 fresh noise images per step at the beginning and 2 per step after the decay. This prevents the buffer from becoming too narrow or collapsing into a small family of patterns.

A fraction 1% of the buffer is also replaced by Fourier textures at each step. These images are generated as sums of random sinusoidal components. They provide global frequency structure and are useful because natural images are known to have non-trivial spectral regularities [16]. Finally, solid-color images are injected during training with a fraction that decreases from 0.5% to 0.05% over the first 10 million images. These random-color injections again act as simple low-frequency seeds rather than semantic content.

After the buffer update, a batch of images is sampled from the buffer for pre-training. The sampled images are horizontally flipped with 50% probability. This flip is applied to the output batch and is not written back into the persistent buffer.

3.3 Why the Buffer Matters

The buffer is more than an implementation detail. It is what makes the generator iterated rather than one-shot. Without a buffer, every generated image would be the result of only a shallow random computation. With a buffer, an image can remain alive for many training steps (40 on average) and be transformed by many different random convolutional networks over time.

A useful way to describe this is as a finite mixture over computation depths:

$$q_{\text{mix}}(x) = \sum_{d \geq 0} \pi_d q_d(x),$$

where q_0 is the initial noise distribution, q_d is the distribution after roughly d rounds of random computation, and π_d is the probability that a sampled buffer image has depth d . In practice, this mixture is controlled by the buffer size, refresh fraction, reset fraction, Fourier injection, random-color injection, and total number of training steps.

This is the main practical link to the universal distribution idea. The true universal distribution mixes over all computable data-generating processes. The IRC buffer instead mixes over a finite family of random convolutional computations and over a finite range of computation depths. Therefore, the approximation is not theoretical in the strict sense. It is an engineering approximation of the same principle: sample many computational sources, iterate them, and hope that generator-specific patterns average out while more general structure remains useful [21].

The buffer also makes the method efficient. Generating a completely new dataset from scratch at every step would be expensive. Keeping one fixed generator would be cheaper, but would make it easy for the target model to learn accidental patterns of that generator. The buffer gives a middle ground: images are reused and transformed over time, while the random computations that act on them keep changing.

4 Training

4.1 Target Model and Pre-training Objective

The target model is a Vision Transformer trained with a masked-autoencoder objective. The final run uses `vit_tiny_patch16_224`. Each input image has resolution 224×224 , and the patch size is 16×16 . This gives

$$N = \left(\frac{224}{16}\right)^2 = 196$$

image patches. Each patch is projected into a 192-dimensional embedding. The encoder has depth 12, 3 attention heads, MLP ratio 4.0, and no dropout. A learnable class token and learnable positional embeddings are used, following the standard ViT design [3].

Since the IRC images have no semantic labels, the pre-training objective cannot rely on class supervision. Instead, this thesis uses masked image modeling, following the general idea of Masked Autoencoders [10]. For each generated image, 75% of the patches are masked. The encoder receives only the visible patches, and a small decoder reconstructs the missing RGB patches. The decoder has embedding dimension 96, depth 2, and 3 attention heads.

Let $\mathcal{M}$ be the set of masked patches. The pre-training loss is

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{M}|} \sum_{j \in \mathcal{M}} \frac{1}{D} \|\hat{p}_j - p_j\|_2^2,$$

where p_j is the original flattened RGB patch, $\hat{p}_j$ is the reconstructed patch, and $D = 16 \cdot 16 \cdot 3$ is the patch dimensionality.

This objective is useful for the present setting because it only requires structure in the input image. If the generated image contains predictable relationships between visible and masked patches, the model can learn from it. If the image is pure independent noise, the reconstruction task should be much less informative. The objective therefore directly tests the central assumption of the thesis: that random computation can turn noise into structure that is useful for learning.

4.2 Training Process

The pre-training loop alternates between generating images and updating the target model. One training step proceeds as follows:

1. The current training progress is passed to the generator, so scheduled values such as update strength, reset fraction, and random-color fraction can be computed.
2. The `irc_conv` generator refreshes part of its persistent buffer using a newly sampled random convolutional network.
3. Fresh noise, Fourier textures, and solid-color images are injected into the buffer according to their configured fractions.
4. A batch of 1000 images is sampled from the buffer and each image is horizontally flipped with a 50% probability.
5. The MAE model masks 75% of the patches and reconstructs the masked RGB patches.
6. The target model parameters are updated with backpropagation. The generator itself is not trained.

The final pre-training configuration is summarized in Table 1.

Component	Configuration
Target model	ViT-Tiny/16, 224×224 input images
Encoder	embedding dimension 192, depth 12, heads 3, MLP ratio 4.0
MAE decoder	embedding dimension 96, depth 2, heads 3
Mask ratio	0.75
Generator	<code>irc_conv</code>
Generator mode	<code>random_generator_per_batch</code>
IRC updates per refresh	$k = 2$
Random convolutional network	width 16, depth 3, kernel size 3, dilation 1, weight std 0.08
Buffer size	20,000 images
Refresh fraction	0.025
Reset fraction	$0.001 \rightarrow 0.0001$ over 10M images
Fourier fraction	0.01
Random-color fraction	$0.005 \rightarrow 0.0005$ over 10M images
Initial random-color fraction	0.10
Update strength	$0.30 \rightarrow 0.15$ over 10M images
Quantization	enabled, 256 levels per channel
Output augmentation	horizontal flip probability 0.5
Training budget	20M generated images
Batch size	1000
Optimizer	AdamW, lr 7.5×10^{-4} , weight decay 0.05
Checkpoints	every 1M generated images
Generator seed	1746

Table 1: Final IRC pre-training configuration used for the main ViT-Tiny experiment.

5 Evaluation

5.1 Downstream Evaluation

After pre-training, the MAE decoder is not used for classification. The encoder is frozen, and features are extracted for each downstream dataset. The main embedding mode used in the final run is `cls_mean`. It concatenates the final class-token representation with the mean-pooled patch-token representation:

$$h_{\text{cls_mean}} = [h_{\text{cls}}; \text{mean}(h_{\text{patches}})].$$

Since the ViT-Tiny encoder has embedding dimension 192, this produces a 384-dimensional feature vector.

A linear classifier is then trained on top of the frozen features. The linear probe is trained for 30 epochs with SGD, learning rate 0.1, momentum 0.9, weight decay 10^{-4} , batch size 8192, and cross-entropy loss. This protocol measures the quality of the representation without allowing the encoder itself to adapt to the downstream task.

In addition to classification accuracy, the evaluation also reports zero-adaptation reconstruction metrics. These are computed using the pre-trained MAE model on real images without further training. The two evaluations answer different questions. The classification task tests whether the frozen encoder has learned a useful global representation of the image: its overall visual character, texture, shape, and structure. In contrast, reconstruction focuses more on local details. It asks whether the model can recover missing patch content, such as exact colours, edges, contrast, and fine texture.

The reported reconstruction metrics are raw-pixel MSE and patch-normalized MSE. For a masked target patch $p_j \in \mathbb{R}^D$, its prediction $\hat{p}_j$, and patch dimensionality $D = 16 \cdot 16 \cdot 3$, raw-pixel

MSE is computed directly in RGB space:

$$\text{MSE}_{\text{raw}}(p_j, \hat{p}_j) = \frac{1}{D} \sum_{d=1}^D (\hat{p}_{j,d} - p_{j,d})^2.$$

This metric is useful because it measures absolute reconstruction fidelity. It checks whether the model predicts the correct RGB values, including brightness, colour, and contrast. A low raw-pixel MSE means that the reconstruction is close to the target patch in the original image scale. However, this metric can be dominated by easy low-frequency information. For example, a model may obtain a relatively low raw-pixel error by predicting smooth average colours, even if it misses sharper local details.

Patch-normalized MSE measures a complementary type of reconstruction quality. For each target patch, the mean and standard deviation are first computed as μ_j and σ_j .

Both the prediction and the target are then normalized using the same target-patch statistics:

$$\tilde{p}_{j,d} = \frac{p_{j,d} - \mu_j}{\sigma_j}, \quad \tilde{\hat{p}}_{j,d} = \frac{\hat{p}_{j,d} - \mu_j}{\sigma_j}.$$

The patch-normalized MSE is then

$$\text{MSE}_{\text{norm}}(p_j, \hat{p}_j) = \frac{1}{D} \sum_{d=1}^D (\tilde{\hat{p}}_{j,d} - \tilde{p}_{j,d})^2.$$

Equivalently, since the same μ_j and σ_j are used for both the prediction and the target, this can be written as

$$\text{MSE}_{\text{norm}}(p_j, \hat{p}_j) = \frac{1}{D} \sum_{d=1}^D \left(\frac{\hat{p}_{j,d} - p_{j,d}}{\sigma_j} \right)^2.$$

Patch-normalized MSE is useful because it measures reconstruction error relative to the local variation of each patch. Instead of only asking whether the model predicts the absolute RGB values, it asks whether the reconstruction captures the patch structure at the patch’s own contrast scale. This makes the metric more sensitive to missing local detail, such as edges and texture, and less dependent on the absolute brightness or colour range of a particular patch.

The final reported reconstruction value is averaged only over masked patches and then over the evaluation dataset:

$$\text{MSE}_{\text{norm}} = \frac{1}{\sum_i |\mathcal{M}_i|} \sum_i \sum_{j \in \mathcal{M}_i} \text{MSE}_{\text{norm}}(p_{i,j}, \hat{p}_{i,j}),$$

where $\mathcal{M}_i$ is the set of masked patches for image i . Using both reconstruction metrics is helpful because they reveal different failure modes. If raw-pixel MSE is low but patch-normalized MSE is high, the model may reconstruct coarse colour and brightness while missing fine structure. If patch-normalized MSE is low but raw-pixel MSE is high, the model may capture local patterns but still make errors in absolute colour or intensity. If both are low, the model reconstructs both the global pixel scale and the local patch details well.

These reconstruction metrics are not a substitute for downstream accuracy. A model can learn to reconstruct pixels without necessarily learning features that are useful for classification. Therefore, classification accuracy and reconstruction error should be read together: classification measures whether the learned representation transfers to real visual recognition, while reconstruction measures how well the MAE objective itself transfers to the detailed structure of real images.

5.2 Datasets and Splits

The downstream datasets are used only after pre-training. During IRC pre-training, the model never sees any real image from these datasets. After pre-training, the encoder is frozen, features are extracted, and a linear classifier is trained on top of these fixed features. In this way, the datasets measure what kind of visual structure the model has learned from the generated IRC images, rather than how well the full model can adapt through end-to-end fine-tuning.

The evaluation suite is split into two groups. The first group is used as a validation group during development. These datasets are useful for comparing generator choices, inspecting training behavior, and understanding which kinds of visual structure are learned. The second group is kept as the final test group. This separation makes the evaluation more reliable, because the final conclusions are not based only on the datasets that were used while choosing the method.

The datasets were chosen to cover several different visual regimes. Some are simple object-recognition sanity checks, such as CIFAR-10. Some are harder or more fine-grained, such as Food-101, Oxford-IIIT Pets, Stanford Cars, and CIFAR-100. Some focus strongly on texture, such as DTD and the synthetic texture tasks. Some test domain shift, such as EuroSAT and RESISC45. ImageNet-1K is included as the main standard reference point, while ImageNet-R tests robustness to style and rendition shifts. Finally, SUN397 tests scene and place recognition, where global layout matters more than recognizing one centered object.

All images are evaluated at resolution 224×224 , matching the ViT input resolution used during pre-training. For small low-resolution datasets, the whole image is resized directly to 224×224 . For larger natural-image datasets, the shorter side is first resized to 256, followed by a center crop of size 224×224 . This keeps evaluation deterministic and avoids introducing extra augmentation during probing.

For the real datasets, the linear probe is trained on the training split and evaluated on the corresponding evaluation or test split. The encoder remains frozen throughout this stage. For ImageNet-R, the task is treated as a robustness evaluation: the probe is trained on ImageNet-1K features and evaluated on ImageNet-R using the overlapping class mapping. This tests whether the ImageNet classifier transfers to style-shifted versions of the same semantic categories.

The synthetic validation datasets are generated only for evaluation. They are not used during IRC pre-training. Each synthetic dataset contains 50,000 training images and 10,000 test images, and the class distribution is balanced. This makes the synthetic tasks useful diagnostic probes: if the model performs well, the result is not caused by class imbalance or by memorizing generated samples from pre-training.

ShapeColorObjects. This dataset contains 8 classes:

{circle, square, triangle, ellipse, star, ring, cross, polygon}.

Each image contains one main foreground object on a random background. The main object can be a circle, square, triangle, ellipse, star, ring, cross, or polygon. Its position, scale, rotation, color, background texture, and partial occlusion are randomized. The classification label is the main object shape.

LineFieldOrientation. This dataset contains 12 classes, formed by 4 orientations and 3 frequencies:

$$4 \times 3 = 12.$$

The orientations are horizontal, vertical, diagonal 45° , and diagonal 135° . The frequencies are low, medium, and high. Each image is made of several irregular regions, where each region contains

Dataset	Type	Preprocessing	Purpose
CIFAR-10 [32]	Real, val	Resize 32×32 to 224×224	Easy object-recognition sanity check. It tests whether the representation captures basic object and texture information.
Tiny-ImageNet [33]	Real, val	Resize 64×64 to 224×224	Medium-difficulty ImageNet-like validation dataset. It checks whether transfer extends beyond very small object benchmarks.
Food-101 [34]	Real, val	Resize short side to 256, center crop 224	Fine-grained food recognition. It is useful because food categories often depend strongly on color, texture, and local visual detail.
Oxford-IIIT Pets [35]	Real, val	Resize short side to 256, center crop 224	Fine-grained animal recognition. It tests whether the representation captures animal shape, fur texture, pose, and background variation.
DTD [36]	Real, val	Resize short side to 256, center crop 224	Texture recognition. This is especially relevant because IRC images often contain texture-like procedural structure.
EuroSAT RGB [37]	Real, val	Resize 64×64 to 224×224	Simple remote-sensing transfer task. It tests whether the learned features transfer away from ordinary object photographs.
ShapeColorObjects	Synthetic, val	Generated at 224×224	Controlled diagnostic for shape, color and object/background separation.
LineFieldOrientation	Synthetic, val	Rendered at high resolution, then antialiased to 224×224	Controlled diagnostic for orientation and frequency sensitivity. It tests whether the encoder captures directional and repeated structure.
TextureMosaic	Synthetic, val	Generated at 224×224	Controlled diagnostic for local texture and region-level structure. It tests whether the representation separates different procedural texture families.

Table 2: Validation datasets used during development and analysis. These datasets are used after pre-training with a frozen encoder and a linear probe.

Dataset	Type	Preprocessing	Purpose
CIFAR-100 [32]	Real, test	Resize 32×32 to 224×224	Harder low-resolution object recognition with many more classes than CIFAR-10. It tests more fine-grained category separation.
ImageNet-1K [4, 5]	Real, test	Resize short side to 256, center crop 224	Main credibility benchmark. It gives the most standard reference point for visual representation quality.
ImageNet-R [38]	Real, test	Resize short side to 256, center crop 224	Out-of-distribution robustness benchmark with renditions, sketches, cartoons, and other style-shifted versions of ImageNet classes.
Stanford Cars [39]	Real, test	Resize short side to 256, center crop 224	Fine-grained recognition of man-made objects. It tests whether the representation separates visually similar classes.
RESISC45 [40]	Real, test	Resize 256×256 to 224×224	Harder remote-sensing scene-recognition task. It tests transfer to aerial imagery and structured scene layouts.
SUN397 [41]	Real, test	Resize short side to 256, center crop 224	Scene and place recognition. It tests whether the representation captures global layout rather than only object-level cues.

Table 3: Held-out test datasets used for the final comparison.

a line field with a specific orientation and frequency. The classification label is the dominant orientation/frequency class in the image.

TextureMosaic. This dataset contains 8 procedural texture classes:

{perlin_noise, fourier_texture, reaction_diffusion,
cellular_automata, fractal_texture, blurred_noise,
dot_texture, wave_texture}.

Each image is a mosaic of irregular regions, and each region is filled with one procedural texture family. The classification label is the dominant texture family in the image. This dataset is useful because it directly tests whether IRC pre-training helps the model represent texture-like and region-level structure.

6 Experiment Results

6.1 Baselines

We compare our generator against four baselines. These baselines are chosen to separate different possible explanations for success.

The first baseline is **random initialization**. In this case, the encoder receives no pre-training. This gives the simplest reference point: if IRC pre-training is useful, it should improve over an untrained ViT encoder.

The second baseline is **Gaussian blur noise**. This baseline starts from random RGB noise and applies Gaussian smoothing. It tests whether local pixel correlation alone is enough to provide a useful pre-training signal.

The third baseline is **Fourier texture**. Fourier textures contain global frequency structure and repeated patterns, but they do not use a persistent buffer or iterated computation. This baseline tests whether frequency structure alone can explain the benefit.

The fourth baseline is a **real-video source**. This baseline uses frames from a single walking video and applies standard image augmentations. The original video is **Walking in Amsterdam** [30], which was first scaled down to 640×360 for storage and decoding efficiency. This baseline is useful as a compact-data reference: a long walking video contains temporal continuity, natural transformations, lighting changes, scene transitions, and many object appearances. Prior work has shown that a single long video can be surprisingly effective for learning visual representations [30].

6.2 Performance

We evaluate the five pre-training sources described above: random initialization, Gaussian blur noise, Fourier texture, real video, and IRC. The x-axis in the plots shows the number of pre-training images seen, in millions. The y-axis shows either frozen linear-probe top-1 accuracy or zero-adaptation reconstruction error. For the random initialization baseline, the line is constant because the encoder is not pre-trained. For the other methods, the encoder is evaluated at different pre-training checkpoints.

The classification results are shown in Figures 1 and 2. IRC pre-training is the strongest method across all classification datasets, both on the validation group and on the held-out test group. The improvement is already visible after only a few million generated images, and in many datasets the curve rises quickly during the first part of training before slowly saturating. The

learned representation therefore transfers to linear classification, rather than only improving the reconstruction objective used during pre-training.

Validation Top-1 Accuracy Across Datasets

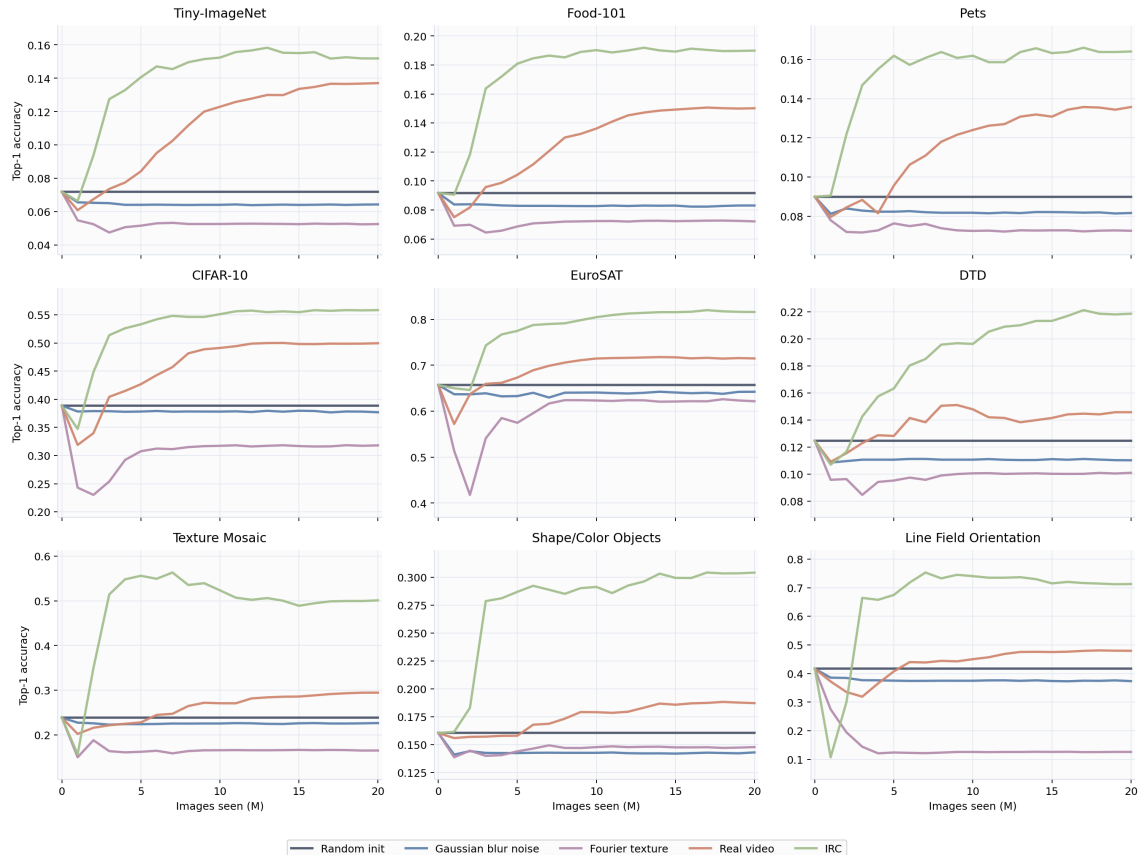


Figure 1: Validation top-1 accuracy across the 9 validation datasets. IRC pre-training gives the strongest frozen linear-probe performance on all datasets. The real-video baseline is usually the second strongest method, while Gaussian blur noise and Fourier texture are much closer to random initialization.

The gain is especially clear for the DTD dataset and controlled synthetic tasks. On LineFieldOrientation, TextureMosaic, and ShapeColorObjects, IRC quickly moves far above the other synthetic baselines. These datasets test simple visual abilities such as orientation, frequency, texture, shape, and region structure. The IRC generator does not generate these datasets directly, and the model does not see their labels during pre-training, so the transfer indicates general visual structure rather than memorization of a particular downstream task.

Test Top-1 Accuracy Across Datasets

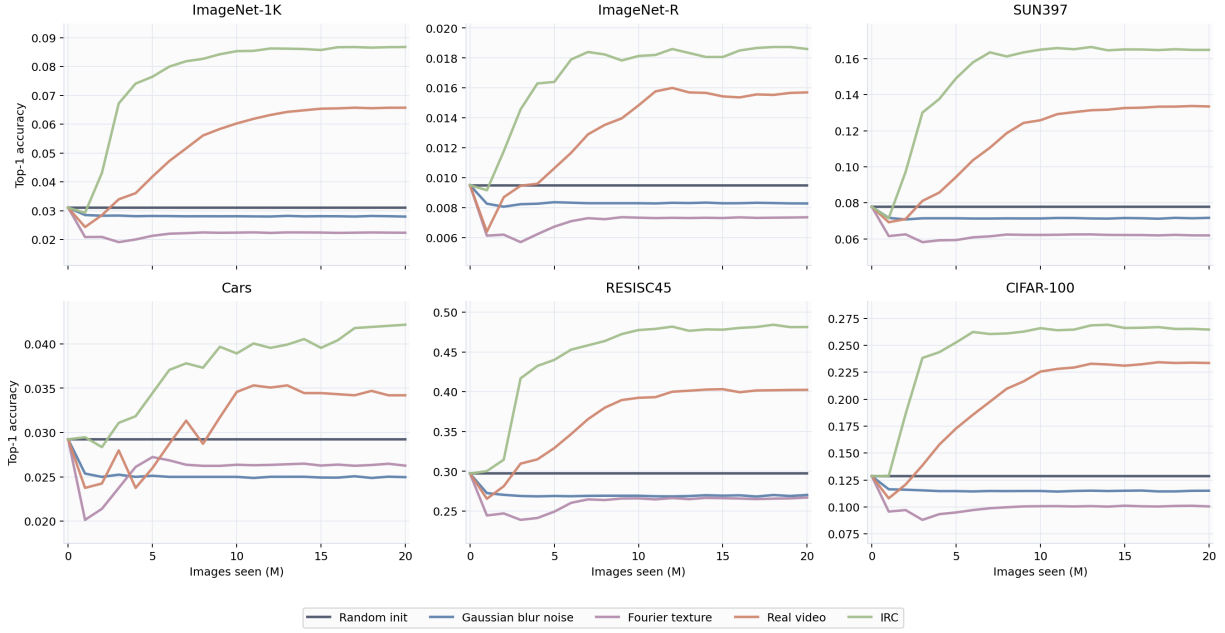


Figure 2: Test top-1 accuracy across the 6 held-out real test datasets. IRC remains the strongest method on the final test group. The absolute numbers are still low compared with large-scale real-image pre-training, but the consistent ordering is important: IRC improves over random initialization, simpler synthetic baselines, and the real video.

The reconstruction plots give a different view of the same models. Classification asks whether the frozen encoder has learned a useful global representation of an image: its general texture, layout, shape, and visual character. Reconstruction asks whether the MAE model can recover missing patch details such as color, brightness, contrast, edges, and fine texture.

Figures 3 and 4 show raw-pixel MSE. This metric measures the direct squared error in RGB space. Any meaningful pre-training reduces reconstruction error compared with random initialization, showing that the MAE objective learns pixel-level structure that transfers beyond the pre-training source.

Validation Raw Pixel MSE Across Datasets

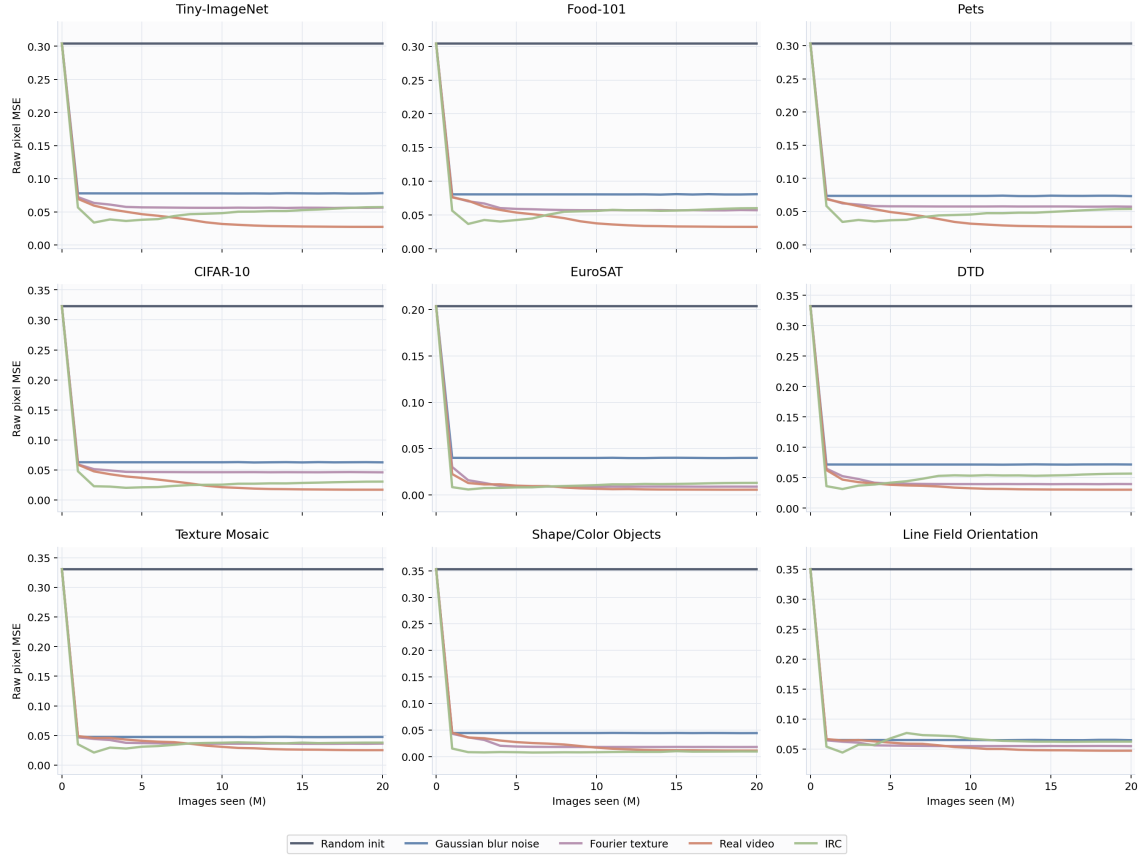


Figure 3: Validation raw-pixel MSE across the 9 validation datasets. All pre-trained methods reduce reconstruction error compared with random initialization. The real-video baseline often obtains the best raw-pixel error on natural images, while IRC remains competitive and clearly better than random initialization.

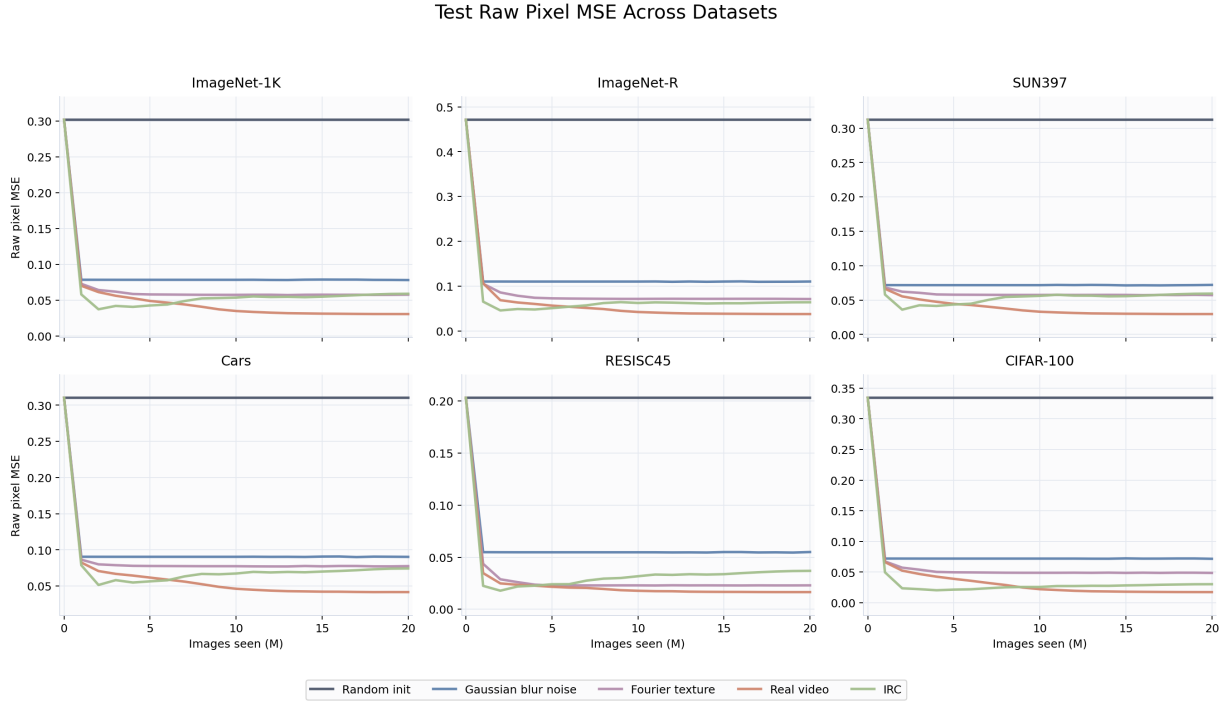


Figure 4: Test raw-pixel MSE across the six held-out test datasets. The reconstruction curves show that pre-training teaches the MAE decoder to reconstruct real images better than random initialization, even when the pre-training data is synthetic.

However, the raw-pixel reconstruction results also show why reconstruction alone is not enough to judge representation quality. The real-video baseline often achieves the lowest raw-pixel MSE, especially on natural-image datasets, because it sees real colors, lighting, and image statistics during pre-training. But this does not always translate into the best classification accuracy. IRC gives better frozen linear-probe accuracy even when its raw reconstruction error is slightly worse. Lower pixel error therefore does not automatically imply better semantic or category-level features.

Figures 5 and 6 show patch-normalized MSE. This metric evaluates reconstruction relative to each patch’s own local mean and variance. Raw-pixel MSE rewards predicting the correct absolute RGB values, including brightness and color. Patch-normalized MSE is more sensitive to whether the model captures local contrast, edges, and texture after removing the easy patch-level scale.

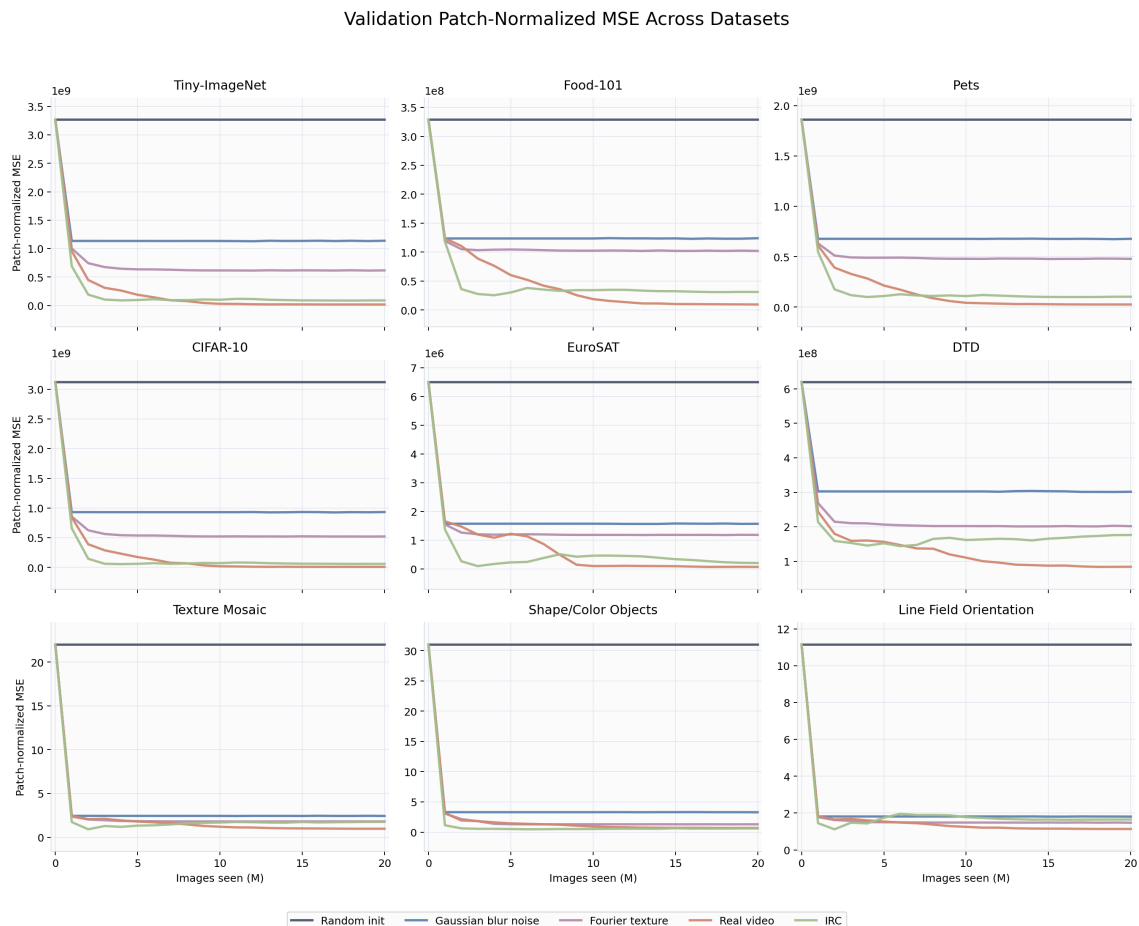


Figure 5: Validation patch-normalized MSE across the nine validation datasets. Patch-normalized MSE removes the mean and scale of each target patch before computing the error, making the metric more sensitive to local structure and contrast-normalized detail.

Test Patch-Normalized MSE Across Datasets

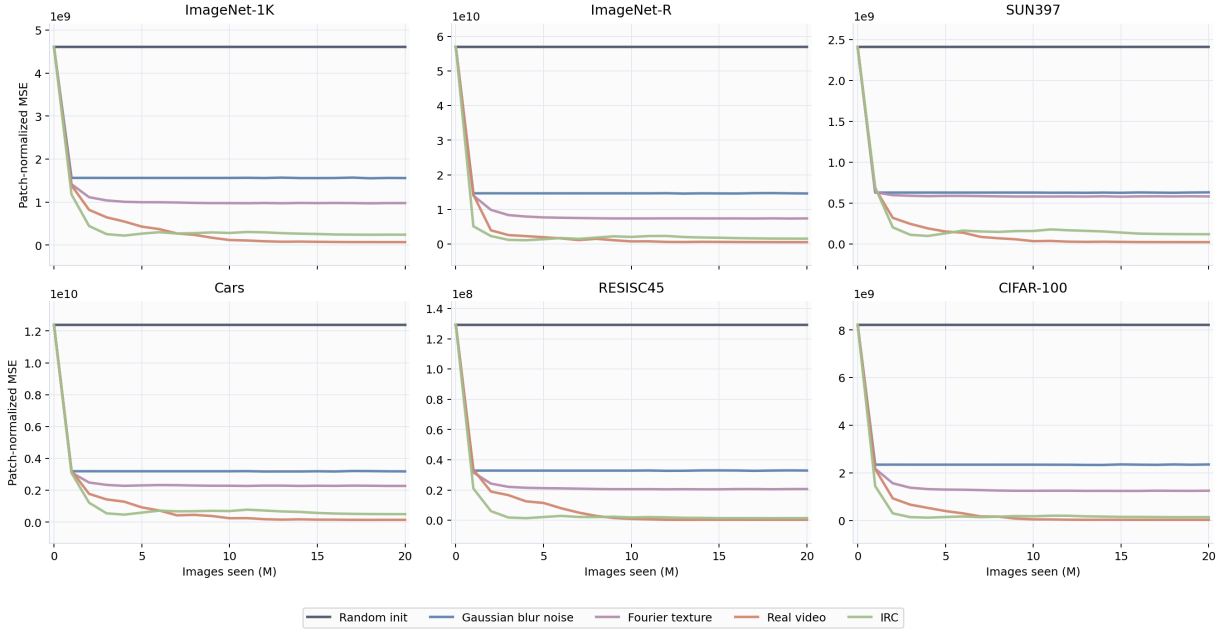


Figure 6: Test patch-normalized MSE across the six held-out test datasets. IRC and real-video pre-training both substantially improve over random initialization and simple noise baselines, but the ordering is not identical to the classification results.

The simpler synthetic baselines are most informative when classification and reconstruction are read together. Gaussian blur noise and Fourier texture focus on narrow low-level image statistics: Gaussian smoothing gives local correlations, while Fourier textures give repeated frequency structure and heavier detail. This helps reconstruction substantially, but it does not produce a strong classification embedding. Gaussian blur noise is usually slightly better for linear classification, while Fourier texture is usually better for reconstruction. This ordering is expected: Fourier provides more detailed local structure for the MAE loss to model, whereas Gaussian blur noise is lighter and less dominated by fine frequency patterns.

Overall, the performance results support three conclusions. First, pure randomness is not enough: Gaussian blur noise and Fourier texture are usually much weaker than IRC for classification. Second, structure matters: IRC improves consistently over simple synthetic baselines, indicating that the iterated random computation and persistent buffer provide a broader learning signal. Third, reconstruction and classification should be interpreted together. IRC does not always produce the lowest reconstruction error, but it produces the strongest frozen representations in the classification evaluation. This is the most relevant result for the research question of this thesis.

7 Limitations and Future Work

The results show that IRC pre-training can produce useful frozen visual representations, but the current study is still an early version of the idea. The main limitation is scale. The final experiment uses a ViT-Tiny model and a pre-training budget of 20 million generated images. This is enough to observe a clear signal, but it is small compared with modern visual pre-training, where both model size and data scale are central to performance [3, 12, 13]. Universal pre-training is also expected

to benefit from scale, because the useful signal comes from averaging over many computational sources and learning patterns that survive across them [21]. A natural next step is therefore to repeat the best IRC configuration on a larger ViT and with scaled hyperparameters.

A second limitation is that the current generator is only a small approximation of the universal-distribution idea. The true universal distribution would mix over all computable data-generating processes. In contrast, `irc_conv` mixes over a much narrower family: small random convolutional networks, a persistent buffer, random resets, Fourier injections, and random-color injections. This is practical and works surprisingly well, but it also biases the generated images toward local texture, smoothness, and frequency structure. This may explain why the method performs well on texture, orientation, remote-sensing, and scene-like tasks, while fine-grained semantic tasks remain difficult. Future generators could include richer random computations, such as cellular automata, reaction-diffusion systems, multi-scale coordinate warps, random attention blocks, compositional shape programs, or simple mechanisms for occlusion and part-whole structure.

A third limitation is the pre-training objective. The current model is trained with an MAE-style masked reconstruction loss on raw RGB patches [10]. This objective is useful because it does not require labels, captions, or real images. However, pixel reconstruction can focus strongly on local detail and may not always produce the best global representation for recognition. The reconstruction results also show that lower pixel error does not necessarily mean better classification accuracy. Future work should therefore test objectives that operate more directly in representation space, such as contrastive learning, DINO-style self-distillation, iBOT-style masked token prediction, or JEPA-style latent prediction. A hybrid objective may be especially useful: reconstruction could preserve local visual detail, while a representation-level loss could encourage more semantic and invariant features.

The evaluation is also limited. Frozen linear probing is clean and easy to compare, but it is only one view of representation quality. The current evaluation does not include end-to-end fine-tuning. A natural next step is to fine-tune the IRC-pre-trained model on a large supervised dataset such as ImageNet-1K, and compare it with the same model trained from scratch. This would test whether IRC pre-training leads to a higher final accuracy, or whether it reaches the same accuracy with fewer real training images or fewer optimization steps.

Finally, the current experiments do not fully isolate every design choice. The best IRC run depends on several hyperparameters: buffer size, refresh fraction, reset schedule, update strength, convolutional depth, Fourier fraction and random-color injection. The results suggest that the persistent buffer and iterated computation matter, but a more complete ablation study is needed to measure the contribution of each component.

Overall, the current results should be read as a proof of concept rather than a finished replacement for real-image pre-training. IRC does not yet learn strong object semantics, and it is far from the performance of large-scale real-image or model-generated pre-training methods [26]. However, it does show that a ViT can learn useful visual structure before seeing real images, using iterated random computations. This supports the main idea of the thesis: some part of learning to see can begin from structure alone, and random computation is one possible way to generate that structure. The open question is how far this idea can be pushed with richer generators, larger models, better objectives, and more realistic downstream evaluations.

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